



Laboratory Work 3

Optimal control design for LTI Plant(LQR)

Lab Goal: *To design an optimal controller for LTI Plant using the Riccati equation*

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Lab Variant: 28

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1. Introduction

In this laboratory work we will design the optimal controller for the LTI plant using the Ricatti equation for the given plant in the variant, The goal in this lab is to use a Linear control regulator (LQR) that minimizes a quadratic cost function.

Here we will use the Positive definite matrix Q that is used to determine the rate of convergence of the states towards zero. And a scalar R that controls the magnitude

$$\dot{x} = Ax + Bu \rightarrow Plant$$

The controller structure for the lab is as $u = -Kx$ where K is the gain we can calculate from the Ricatti equation

$$A^T P + PA + Q - Pbr^{-1}b^T P = 0$$

$$K = r^T b^{-1} P$$

Where the State Matrices are,

$$A = \begin{bmatrix} 0 & 1 \\ 4 & -5 \end{bmatrix}, \quad B = \begin{bmatrix} 9 \\ 2 \end{bmatrix}$$

While the Q matrix is

$$Q = \begin{bmatrix} 9 & 0 \\ 0 & 7 \end{bmatrix}, \quad R = 1$$

2. Modeling

We will now model the system on matlab simulink and calculate the gain of the controller using the linear quadratic controller function in matlab

Model initialization function:

```
1 A = [0 1; 4 -5];  
2 b = [9;2];  
3 Q = [9 0; 0 7];  
4 r = 1;  
5 K = lqr(A,b,Q,r);
```

Figure 1. Calculating K

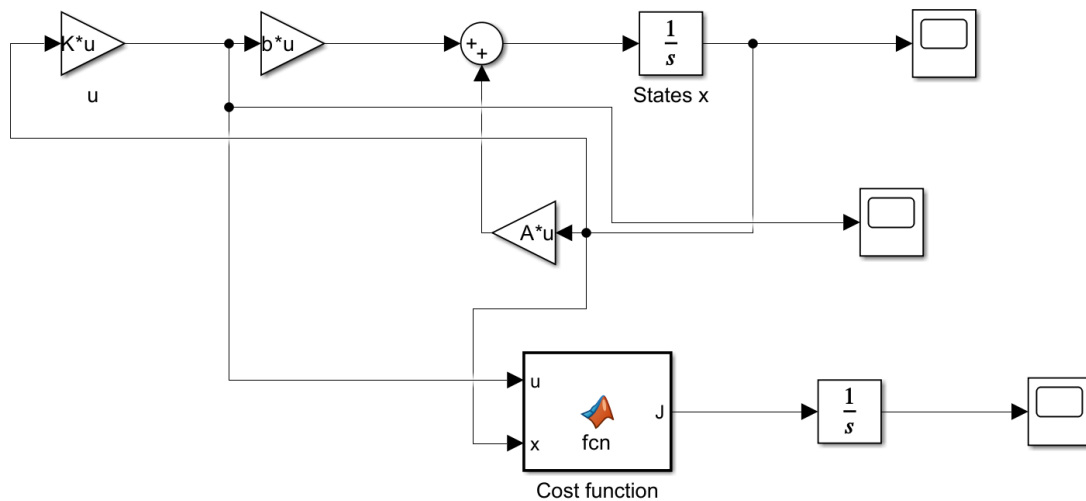


Figure 2. Simulink Model of the system

We will now simulate the system with the given initial conditions as $x(0) = [1,0]^T$

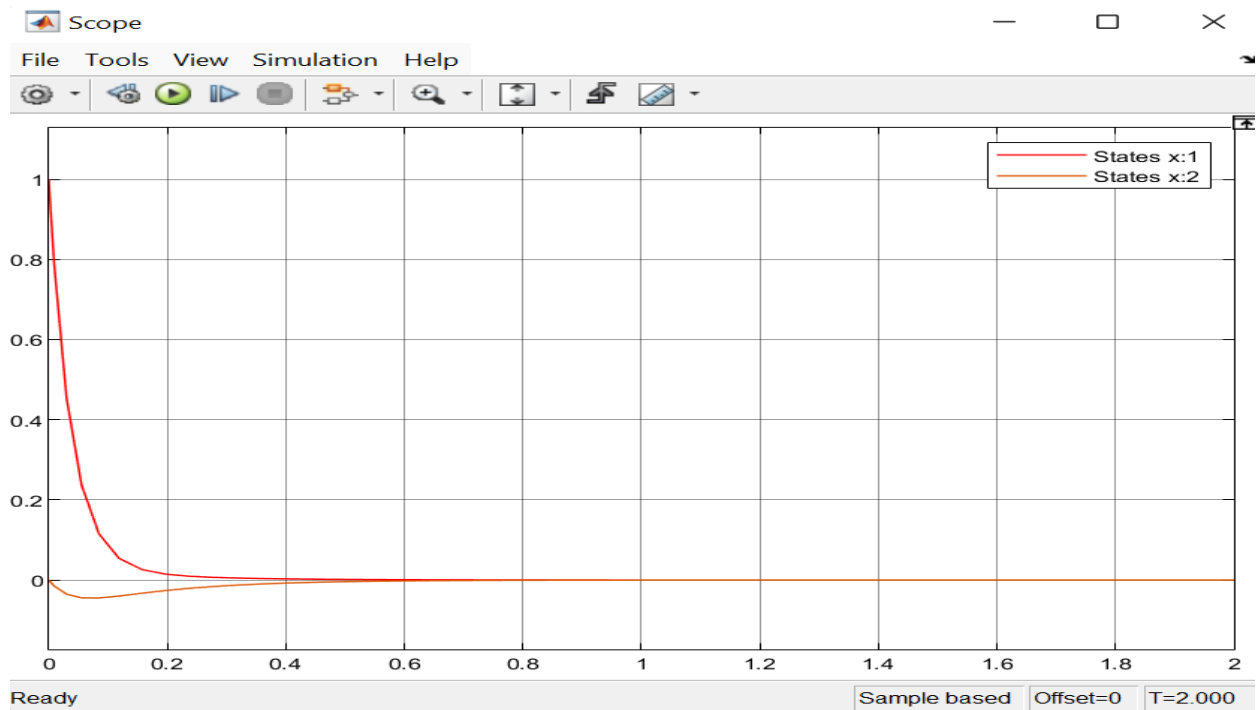


Figure 3. States x1 and x2

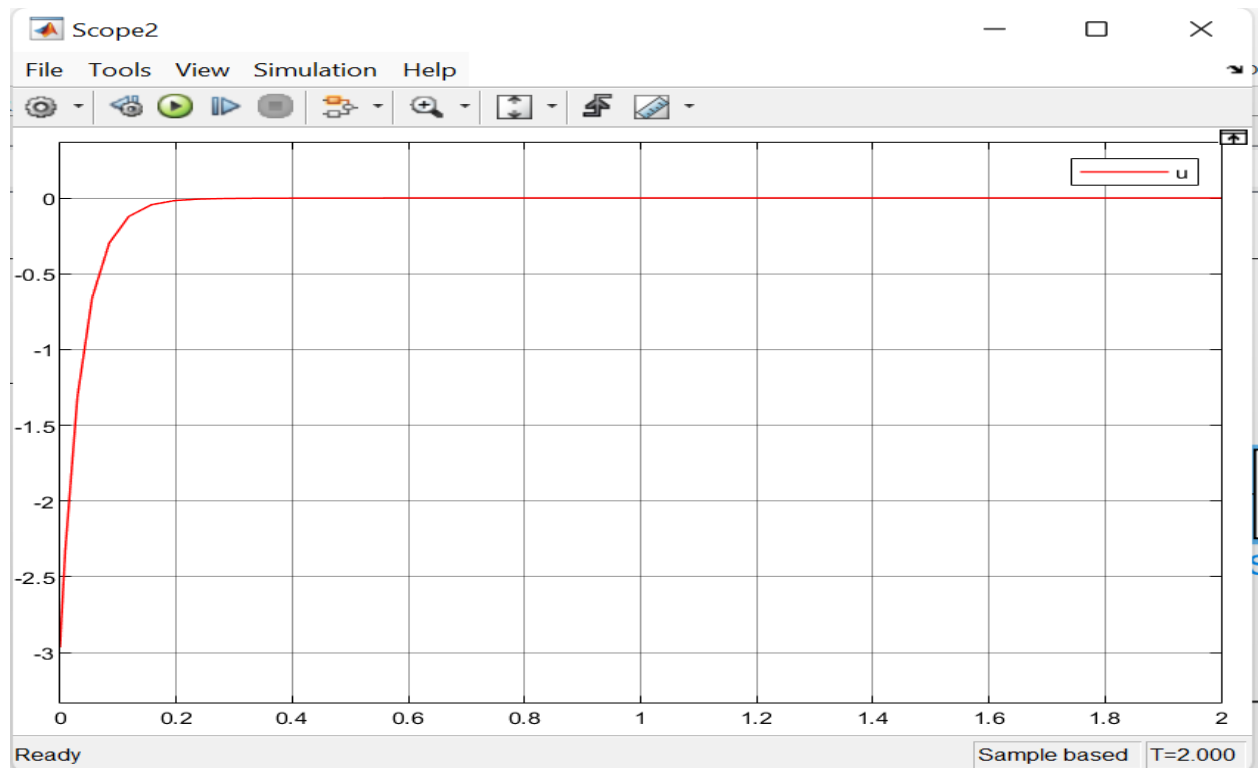


Figure 4. Control signal u

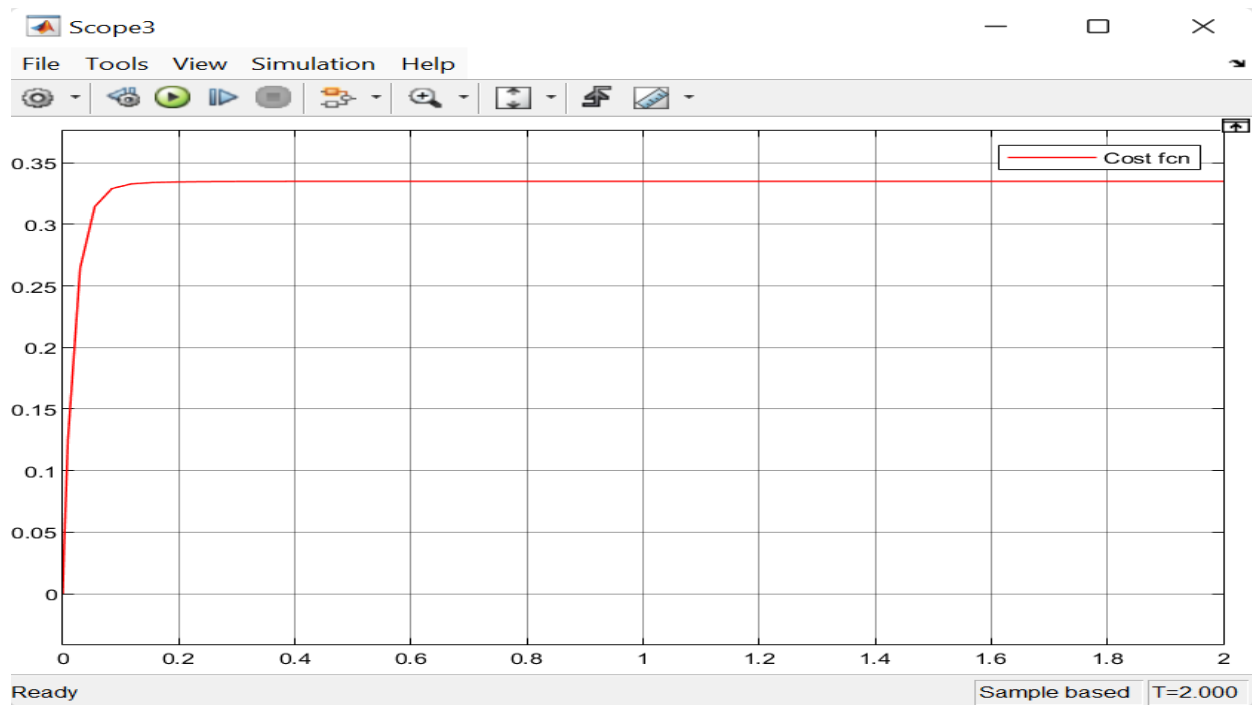


Figure 5. Cost function

Now we will negligibly change the value of K and then observe the resulting plots and the cost function,

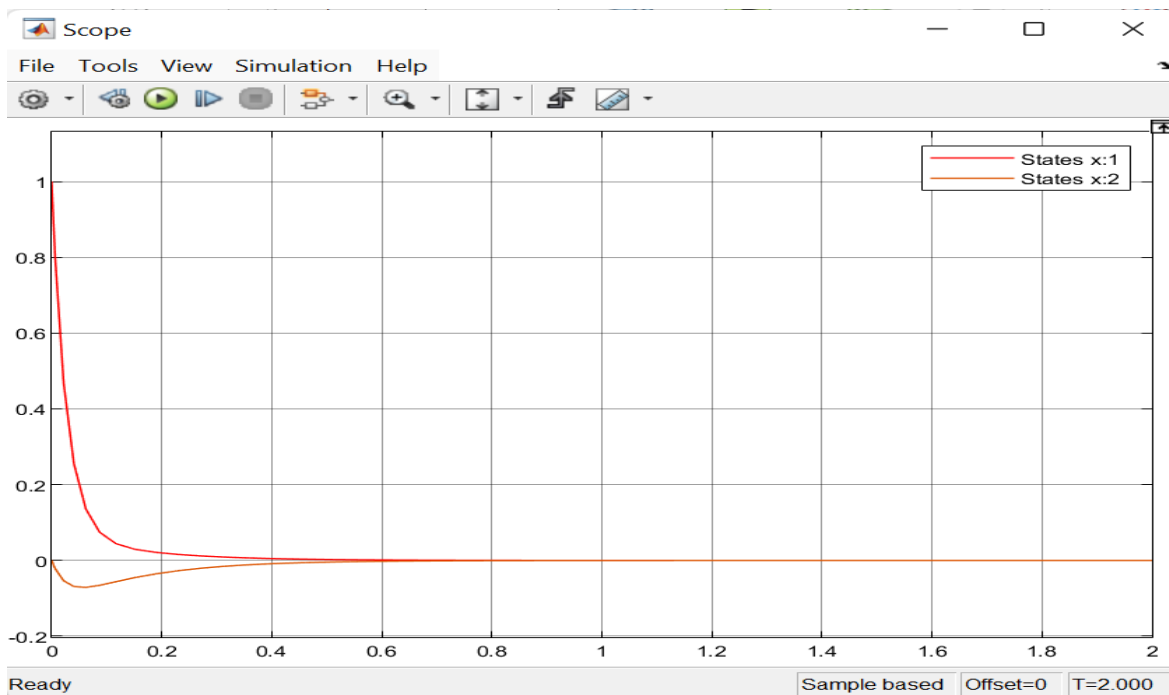


Figure 6. States x_1 and x_2 (Changing K)

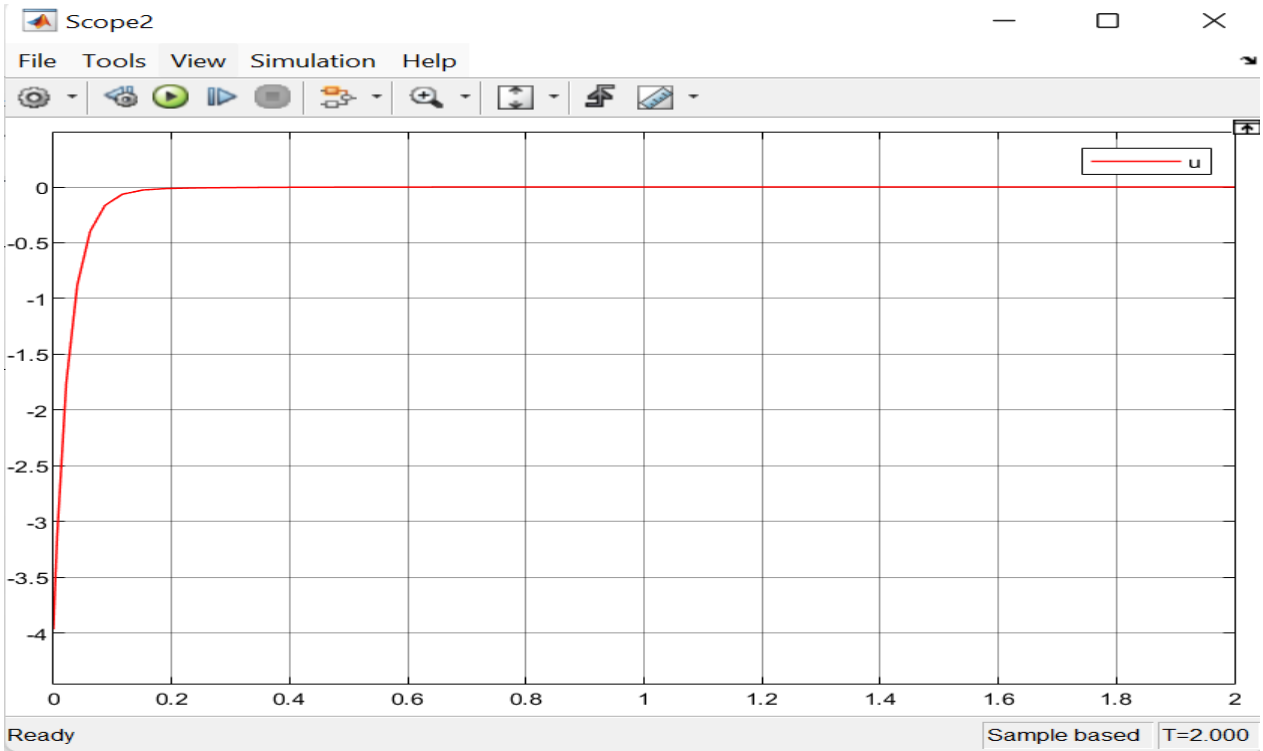


Figure 7. Control signal u (Changing K)

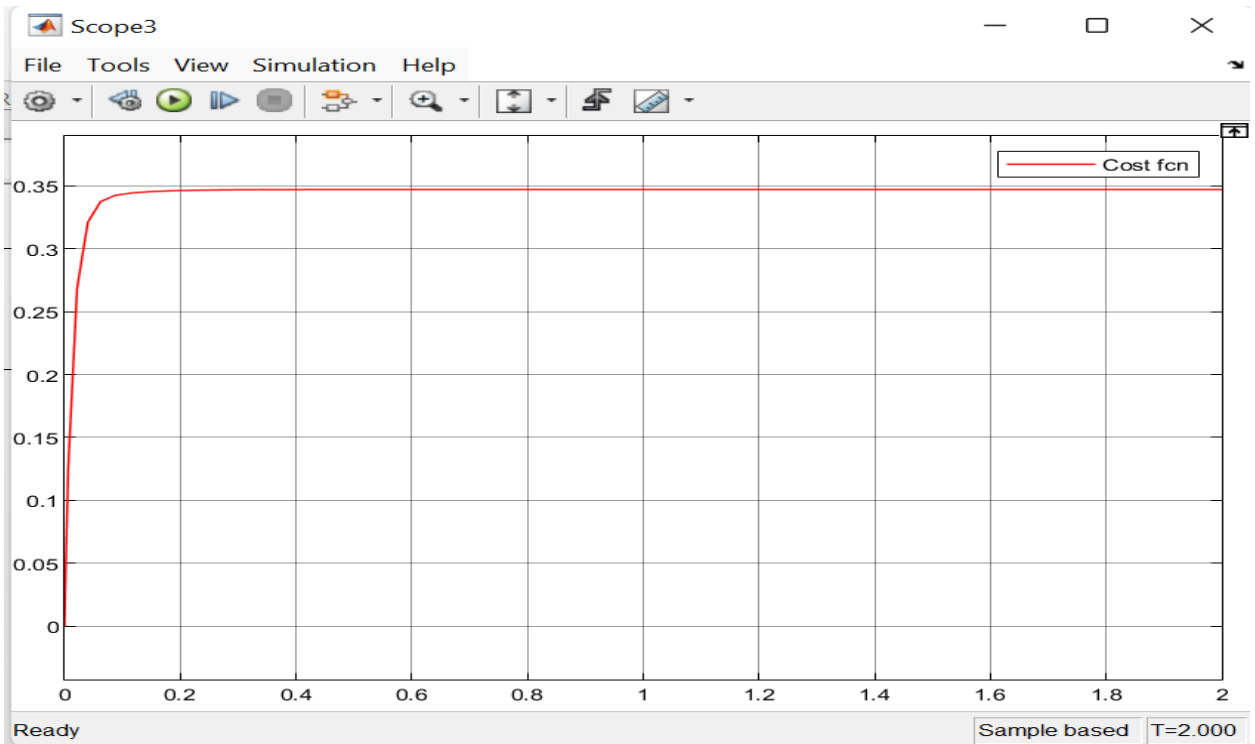


Figure 8. Cost function (Changing K)

By negligibly changing the value of K we can observe that the system is taking a bit more time to converge to zero and the cost function has slightly increased as compared to the optimal solution.

Now we will simulate the system on three different coefficients r and Q with the conditions if $r > 0$, then $Q = kQ^*$ where k is a positive gain and the Q^* is the original Q matrix.

For this purpose we will use

$$r_1 = 5, r_2 = 10, r_3 = 15 \text{ and } Q_1 = 2 * Q, Q_2 = 5 * Q, Q_3 = 10 * Q$$

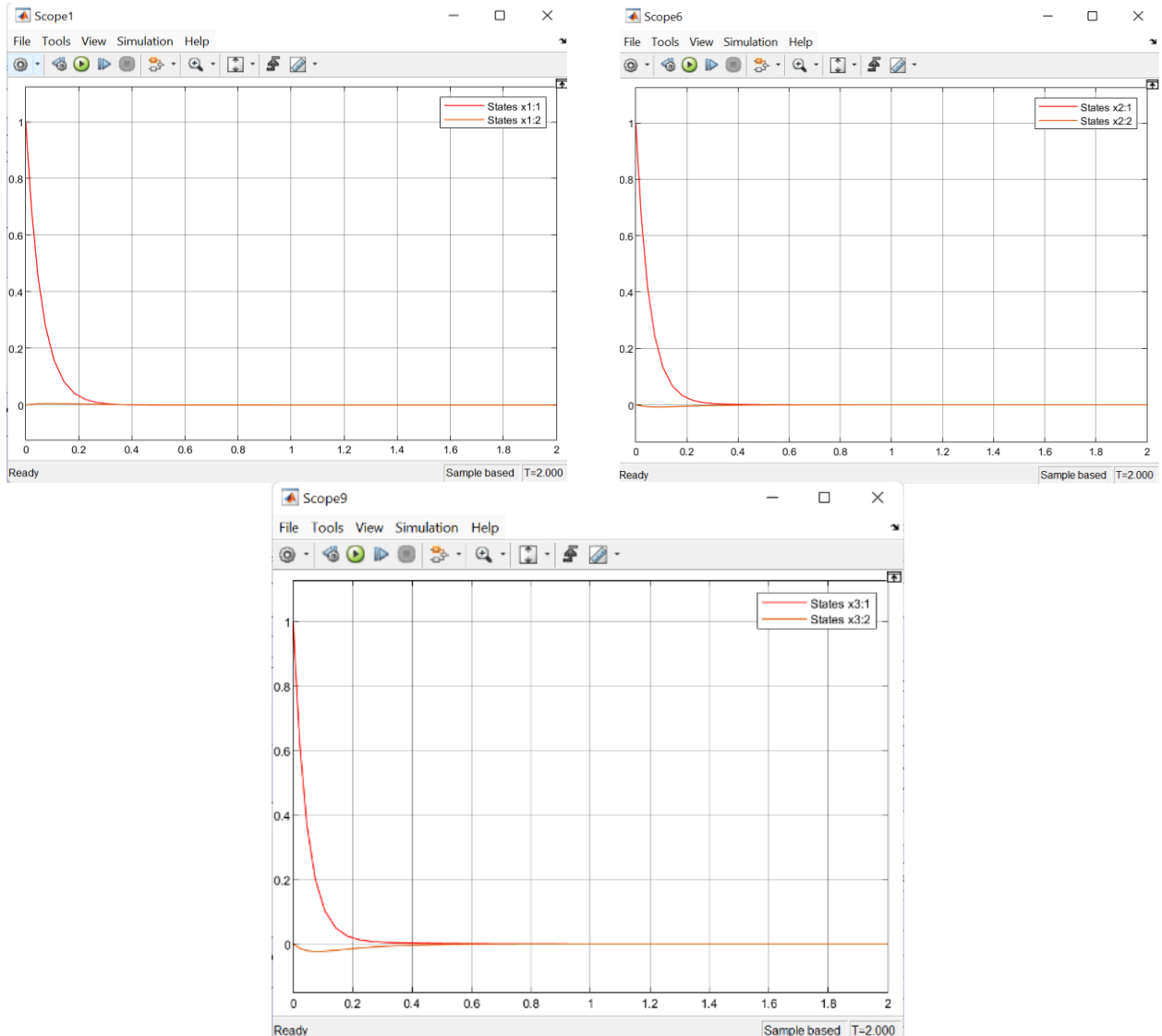


Figure 9. Comparison between 3 states on different r and Q

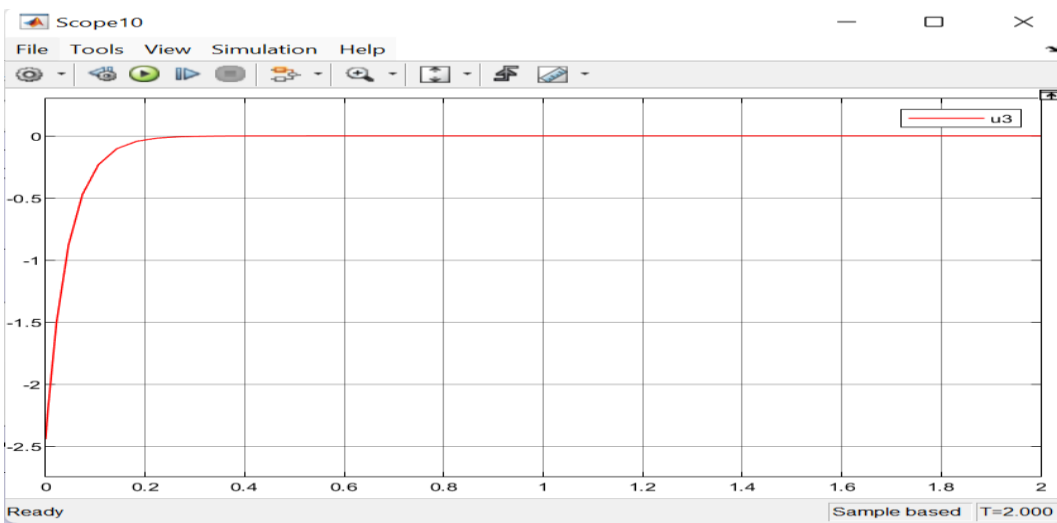
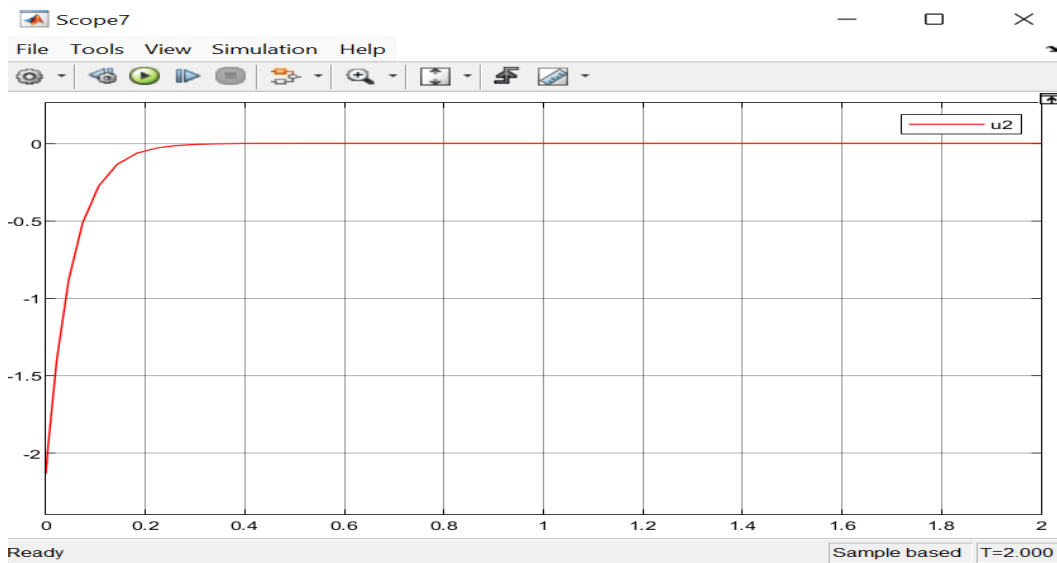
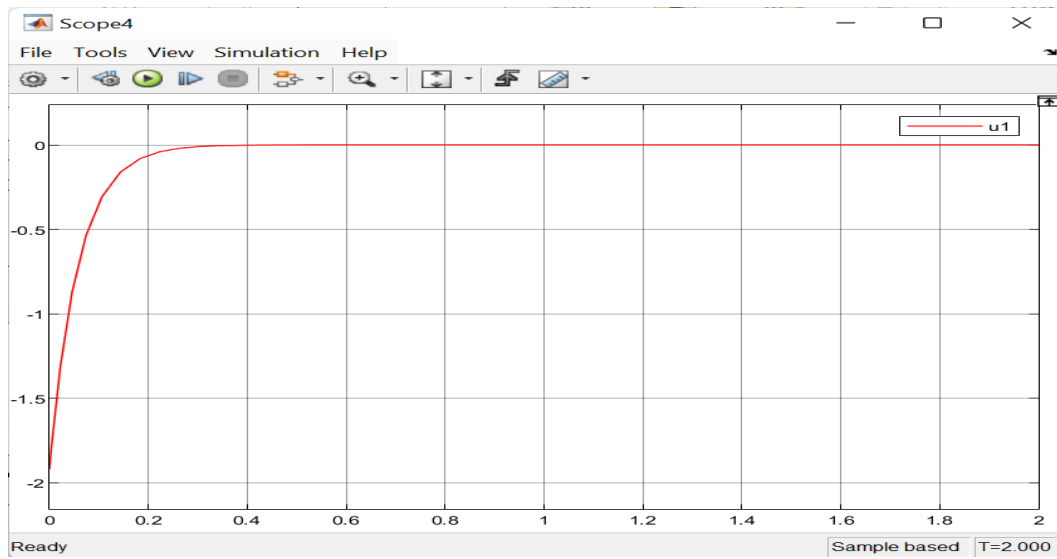


Figure 10. Comparison between 3 control signals on different r and Q

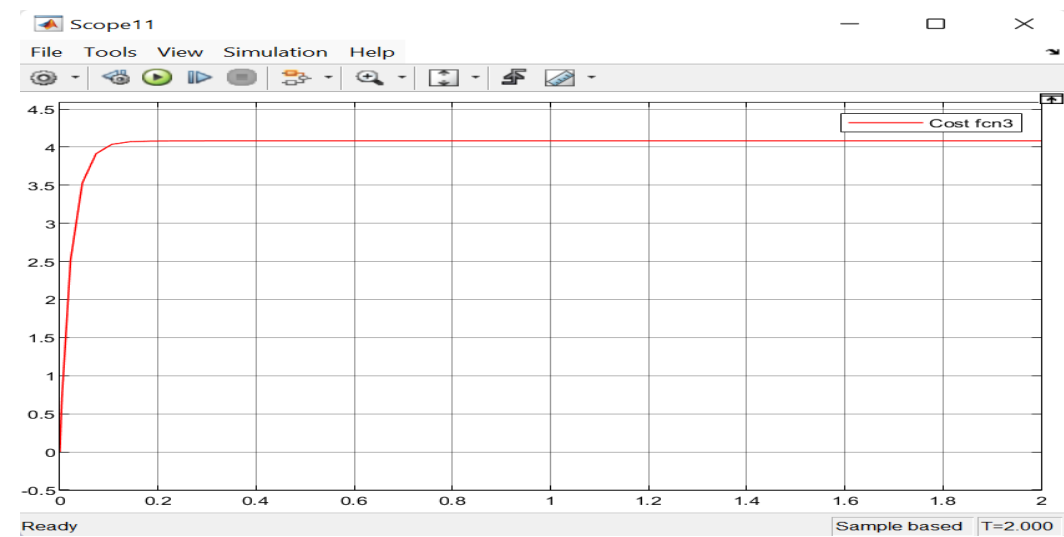
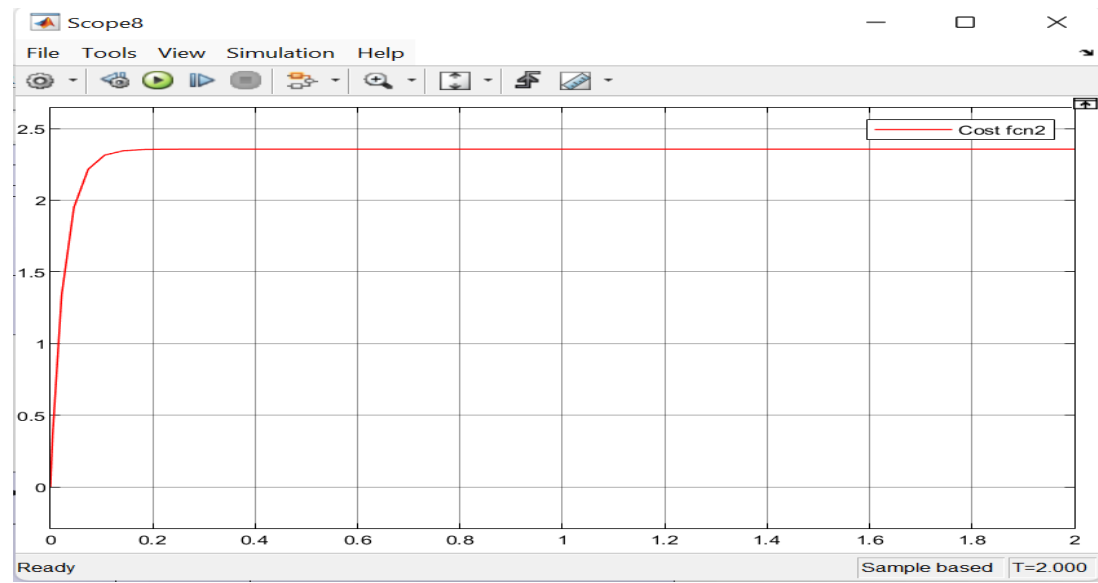
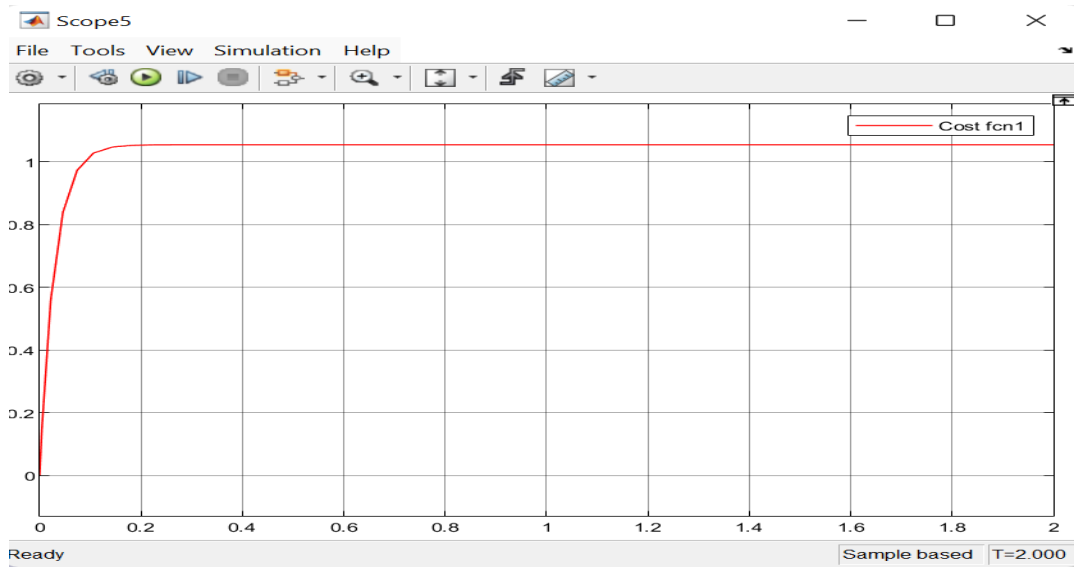


Figure 11. Comparison between cost function on different r and Q

3. Conclusion

In this laboratory we designed and obtained the controller parameters using the Riccatti equation on our given State space matrices, coefficient r and the Matrix Q ,

We have observed that by changing the value of Q and r it affects the convergence of the output of the system. On a higher value of r and Q the system converges with a higher cost function and the control signal starts to converge from a lower value, nevertheless the states are converging to 0 in all the cases but the cost is different in all cases and much higher than the optimal controller.